

SETHULAKSHMI BABU

M2 in Data and AI | Seeking interested internships from Feb 2026

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EDUCATION

- Master's in engineering - Major: Data and AI | ongoing | ESILV Engineering School, Courbevoie
- Bachelor's in computer science | 2023 | Mahatma Gandhi University, Kottayam | 88.80%

ACADEMIC PROJECTS

- **Credit Card Fraud Detection** – Deployed real-time ML pipelines using **XGBoost/AdaBoost**; **designed custom evaluation metrics** to maximize detection rate by focusing on cost-sensitive false negatives.
 - **Logistics Optimization System** – Developed AI-driven path optimization with enhanced A* algorithm, dynamic penalties, and real-time API integration for adaptive route generation.
 - **Prompt Engineering (Vertex AI)** – Engineered production-ready prompt pipelines on **Vertex AI** utilizing advanced Chain-of-Thought reasoning and function calling to enhance consistency and reliability of LLM outputs.
 - **Clinical Diagnosis AI** – Developed a Random Forest model for infectious disease triage, strategically engineered for **high-sensitivity performance** to minimize high-stakes diagnostic errors. Demonstrated expertise in model integrity by rigorously applying enhanced preprocessing and strictly **preventing data leakage**.
 - **Drone Health Guard** – Implemented an end-to-end ML pipeline using **Random Forest** for real-time sensor-level fault detection, enabling **predictive maintenance** through feature ranking.
- Hackathon participation:** Presented this project at [Hackathon held by ESILV and IBM](#)

EXPERIENCE

Independent Author (Self-Published) | 2017–Present

- Managed full lifecycle production of multiple long-term creative projects, demonstrating high-level **autonomy**, resource planning, and **long-term project commitment**.

CERTIFICATIONS

- [IBM watsonx.data Technical Essentials](#) | 2025
- [Getting Started with Generative AI](#) | 2025
- Enablers Agile Methodology | 2024
- Ethical hacking | 2023 | Avodha Institute

SKILLS

Time Series & Forecasting: Time series analysis, anomaly detection

Machine Learning & AI: Random Forest, XGBoost, Logistic Regression, Neural Networks, CNNs, feature engineering, model evaluation, error analysis, outlier detection, LLM fine-tuning, prompt engineering

Deep Learning & LLMs: TensorFlow, Keras, LLM-based text generation, image generation

Programming: Python, SQL (MySQL / PostgreSQL / Oracle), JavaScript, TypeScript, Node.js (Express), C++, Java

Data Analytics & Statistical Skills: Statistical analysis, root-cause analysis, data preprocessing, EDA

Frontend: React, HTML, CSS

Soft Skills: Analytical, proactive, detail-oriented, autonomous, curious, problem-solver

LANGUAGES

French – B1 | English – C1 | Malayalam – Native